Richard Yu's Curriculum Vitae.

Contact Information

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Current Scientific Interests

To study complex molecular biological systems by developing computational and experimental technologies to 1) accurately map molecular networks, 2) monitor the flow of information through them, and 3) interpret and model these data.

Education and Research Experience

9/2000 − present
: Research Fellow, The Molecular Sciences Institute, Berkeley, CA.
Developing sensor technologies for measuring information flow in cellular signaling pathways at single−cell resolution.

1/1995 − 5/2000
: Graduate Student, Laboratory of Dr. Axel T. Brunger, Department of Molecular Biophysics and Biochemistry, Yale University, New Haven, CT
Solved and analyzed the high−resolution crystal structures of two domains of NSF, a hexameric ATPase involved in eukaryotic membrane fusion. Formulated plausible mechanisms for ATP−dependent hexamerization and ATP hydrolysis−dependent conformational changes. Ph.D. in Molecular Biophysics and Biochemistry, May 2000.

5/1993 − 6/1995
: Research Associate, Laboratory of Dr. Teresa Head−Gordon, Life Sciences Division, Lawrence Berkeley Laboratories, Berkeley, CA
Designed and implemented an improved feed−forward back−propagation artificial neural network (in C) for protein secondary structure prediction. By hard−coding physiochemical information about sequence−structure relationships, modest gains in prediction accuracy were achieved.

8/1989 − 5/1993
: Undergraduate Student, Laboratory of Dr. Robert M. Glaeser, Department of Molecular Cell Biology, University of California, Berkeley, CA
Attempted 2D crystallization of the bacterial histidine transport membrane protein complex. B.A. (Honors) in Molecular Cell Biology, emphasis in Biophysics, May 1993. Minor in Computer Science, emphases in Graphics and Artificial Intelligence

Skills and related experience

Molecular Biology and Biochemistry

◊ Standard DNA/RNA techniques: cloning, PCR, blots.
◊ Standard bacteria and yeast lab techniques.
◊ Protein expression: native sources, bacterial.
◊ Protein purification: HPLC/FPLC.
◊ Protein engineering: mutation, domain rearrangement/swapping, molecular evolution.

◊ Protein analysis: gel−based techniques (SDS, Western, IEF), analytical equilibrium ultracentrifugation, multi−angle laser light scattering, intrinsic fluorescence spectroscopy, circular dichroism spectroscopy.
◊ DNA and protein bioinformatics: standard Web−based services (alignments, property predictions) and databases, Staden, Phred/Phrap/Consed.

Protein Crystallography

◊ Crystal preparation: vapor diffusion and batch crystallization, dialysis.
◊ Data collection: capillary mounting, cryoprotection and cryo−mounting, rotating−anode and synchotron radiation sources with various detectors (R−AXIS IV, MacScience, and MAR; ADSC CCD. Synchotron data collection at ALS line 5.0.2, SSRL line 1−5, BNL lines X4A and X12.
◊ Data processing: HKL Suite (Denzo, Scalepack), DREAR.
◊ Model building: O and Uppsala Software Factory programs.
◊ Refinement: X−PLOR and CNS
◊ Structure analyses: GRASP, Surfnet, DALI, SC
◊ Molecular graphics software: PyMol, GL_render, Molscript/Bobscript, Raster3D, POV−Ray.

Computer Skills

◊ Programming languages [1 (marginal) to 3 (proficient)]: C(3), C++(1), Lisp (3), Perl (2), awk/sed (3), Python (1), Java (1), HTML (3), OpenGL (3), Tcl/Tk (1), basic relational database usage (1). Have been programming since well before my college years, though for last few years have only been primarily scripting languages (Perl/Awk). Can quickly learn languages and systems.
◊ Operating systems/system administration: *nix (mosty Linux and AIX), Windows 95/98/NT, DOS, MacOS

Other

◊ Basic plastic and metal fabrication techniques: MIG/oxy−acetylene welding, precision machining.
◊ Introductory knowledge of topics in neurobiology and electrical engineering. Strong math and physics background.

Teaching experience

8/1999 − 12/1999 and 1/1997 − 5/1997

Teaching assistant, Molecular Foundations of Medicine
Discussion sections, grading. Expanded and maintained course Web page. Moderated online discussion/Q and A chatroom.

8/1995 − 12/1995

Teaching assistant, laboratory section, Introduction to Biochemistry
Grading. Introduced students to basic biochemical techniques and associated equipment required for the generation of a protein point mutant. Taught computer modelling of modified protein using WhatIf. Delivered lecture on computer graphics and computer modeling of proteins.

First Author Publications

- **Yu, R.C.**, Jahn, R., and Brunger, A.T. NSF N−terminal domain crystal structure: models of NSF function. *Mol. Cell* 4, 97−107 (1999).
- **Yu, R.C.**, Hanson, P.I., Jahn, R. and Brunger, A.T. Structure of the ATP−dependent oligomerization domain of *N*−ethylmaleimide sensitive factor complexed with ATP. *Nat. Struct. Biol.* 5, 803−11 (1998).

- **Yu, R.C.** and Head−Gordon, T. Neural−network design applied to protein secondary structure predictions. *Phys. Rev. E* 51, 3619−3627 (1995).

References

- Dr. Axel Brunger, Department of Molecular and Cellular Physiology, Stanford University, 1201 Welch Rd. P210 MSLS, Stanford, CA 94305−5489. (650) 736−1031, axel.brunger@stanford.edu
- Dr. Reinhard Jahn, Department of Neurobiology, Max−Planck−Institute for Biophysical Chemistry, D−37077 Gottingen, Germany, Am Fassberg. x49−551−201−1634, rjahn@gwdg.de
- Dr. William Summers, Department of Molecular Biophysics and Biochemistry and Therapeutic Radiology, Yale University, 370 HRT, New Haven, CT 06511. (203) 785−2980, william.summers@yale.edu

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